

CHEN ZHIWEI

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RESEARCH INTERESTS

My research interests lie in understanding and developing effective, efficient, reliable, and safe AI systems. My recent work focuses on strategic reasoning for LLM agents and efficient multi-agent inference.

EDUCATION

National University of Singapore Master of Technology in Artificial Intelligence Systems, GPA: 3.51/4.0	Aug 2023 - Nov 2024
Sun Yat-sen University Bachelor of Engineering in Traffic Engineering, GPA: 3.5/4.0	Sep 2018 - Jun 2022

RESEARCH EXPERIENCE

The Hong Kong University of Science and Technology (Guangzhou) Research Assistant Advisor: Prof. Tianxiang Zhao	Apr 2026 - Present
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(Ongoing) Efficient Multi-Agent System via Cross-Agent KV Cache Reuse

- Investigating potential scenarios for cross-agent KV reuse to reduce redundant prefill computation in LLM multi-agent systems.
- Characterized cache reusability across agents with varying prompt lengths and content, and evaluating trade-offs among cache reusability, inference efficiency, and generation quality.

Adaptive Strategic Reasoning for LLM Agents in Imperfect-Information Games

- Investigated adaptive decision-making of LLM agents in repeated imperfect-information interactions, focusing on challenges in strategic reasoning, opponent adaptation, and effective use of historical feedback.
- Proposed SAGE, a training-free strategic reasoning framework that integrates equilibrium-based policies, opponent belief modeling, and memory-based guidance without updating the underlying LLM parameters.
- Evaluated SAGE on Leduc Hold'em, Liar's Dice, and Goofspiel against six game-specific opponent types in each environment, consistently outperforming existing LLM-based game-playing agents.

Center for Artificial Intelligence and Robotics, Chinese Academy of Sciences, Hong Kong

Research Intern → Research Assistant Advisor: Prof. Mingyang Zhao	Mar 2024 - Feb 2026
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- Built data acquisition and PDF-processing pipelines for multimodal dataset construction and retrieval-augmented generation (RAG).
- Evaluated multimodal models on medical visual question answering and visual reasoning tasks.
- Conducted multimodal representation learning research on ultrasound image captioning and image-text alignment.

PUBLICATIONS

- Z. Rao, P. Zhou, D. Chong, **Z. Chen**, S. Wang, N. Tang. "Structural and Disentangled Adaptation of Large Vision Language Models for Multimodal Recommendation". *SIGIR*, 2026

MANUSCRIPTS

- Z. Chen**, T. Wang, T. Zhao. "Adaptive Strategic Reasoning for LLM Agents in Imperfect-Information Games". Submitted to *ICLR* 2027